

**AYON: WEB-BASED INTEGRATED RESEARCH SERVICES MANAGEMENT
SYSTEM FOR MAMITUA SABER INSTITUTE OF RESEARCH AND CREATION
(MSIRC) - MSU MAIN**

A Capstone Project 1
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CHAPTER 1

Introduction

A web-based research management system is an online platform designed to organize, monitor, and manage research activities electronically. It facilitates submission of proposals, requests grants, monitors progress, and retrieves completed research from a single system. Research is an ongoing systematic process of discovering new things, verifying existing data, and resolving issues through systematic investigation and study, (Leedy and Ormrod, 2020). In the Philippines, most universities are increasingly adopting digital tools to make research more accessible and efficient.

In the Mindanao State University – Main Campus, the Mamitua Saber Institute of Research and Creation (MSIRC) is the office that deals with the management of the research activity of faculty, staff. MSIRC handles proposals submissions, travel and publication grants, research review, approval, and archiving complete research papers. These processes are still manually, which leads to delays, loss of data, and challenges in monitoring research progress. To address these issues, the researchers propose a Web-Based Research Management System to automate MSIRC's workflow. Through this system, MSIRC will enhance efficiency, reduce paperwork, and improve communication and transparency among administrators, researchers, and The Institution. As Hernandez (2021) explained, universities without an integrated research management system experience slower submissions and higher chances of data loss, which makes digital transformation necessary.

Previous studies support the value of digital research management systems, yet gaps remain. For instance, Malaya et al. 's (2022) IMSR system at DMMMSU South La Union Campus demonstrates improvements in managing submissions and monitoring research operations, but it does not clearly emphasize modules for comprehensive grant monitoring and long-term archiving in its published description. Broader studies such as Advancing Research Management in the Philippines (2024) also highlight persistent documentation and data management challenges in many institutions. These works collectively show that while partial systems exist, few address the full lifecycle of research in a way that is tailored to institutional workflows. The proposed system aims to fill this gap by integrating proposal submission, grant management, progress monitoring, and archiving into one accessible platform, thereby promoting transparency, efficiency, and sustained research development at MSIRC.

Additionally, the proposed system aligns with the global trend toward digital transformation in higher education, especially in the field of research administration. Many universities worldwide have already shifted from paper-based workflows to automated systems, leading to faster communication, better collaboration and improved data security. Implementing such innovation at

MSU will not only simplify administrative processes but also foster a culture of research excellence and accountability, which is crucial for academic progress and institutional reputation.

1.1 Project Context

The capstone project, “AYON: Web- Based Integration Services Research Management System for Mamitua Saber Institute of Research And Creation (MSRIC) - MSU MAIN,” is a software development initiative focused on designing and developing a centralized, web-based platform for managing research-related processes. The system is intended to automate the submission, monitoring, and documentation of research proposals and projects conducted under MSIRC. The system will specifically be implemented at Mindanao State University – Main Campus, particularly within the Mamitua Saber Institute of Research and Creation (MSIRC). The core concern of the project is to address the limitations of the current paper-based and manual research management system used by the institute. The reliance on traditional processes has been identified as a research gap that results in the following issues: Delays in proposal submission, review, and approval, Risk of lost, misplaced, or incomplete research documents, Difficulty in monitoring research progress and timelines, Inefficient generation of research reports and records

Therefore, to address these challenges, the project “AYON: Web-Based Research Management System for MSIRC” is proposed. It is a web-based system designed to: Allow researchers to submit proposals and required documents online, Enable administrators to review, approve, and manage research projects efficiently, Provide digital archiving of research records for secure storage and retrieval, Improve transparency, organization, and communication within the research community. The system provides organized and up-to-date research data that support institutional monitoring, reporting, and decision-making. Furthermore, AYON contributes to the university’s goal of promoting innovation, accountability, and academic excellence by ensuring a more efficient and transparent research management process.

The key stakeholders who will benefit from this project include: MSIRC administrators and staff, Faculty members and researchers of MSU–Main Campus, University administrators involved in research oversight

Regarding development and implementation, the system will be developed as a web-based platform using technologies such as Visual Studio Code, MySQL, React js and Figma. The conceptualization of the project began during the researchers’ capstone course and is expected to be completed within the academic timeline of the program.

1.2 Purpose and Description

This project entitled “AYON: Web- Based Integration Services Research Management System for Mamitua Saber Institute of Research And Creation (MSRIC) - MSU MAIN,” is intended to digitize and improve the management of research-related processes within the university. The system aims to replace the existing manual submission, review, and approval procedures with a more efficient, organized, and automated web-based platform.

The automation of research workflows is the primary focus of this project, addressing issues such as time-consuming paperwork, processing delays, and inconsistent record handling. Through AYON, the researchers aim to strengthen research monitoring, documentation, and communication between MSIRC and the university’s research community.

Specifically, the system is designed to provide:

- A secure web-based platform for online research proposal submission
- A digital process for reviewing, approving, and tracking research projects
- An organized repository for storing and retrieving research documents
- An efficient communication channel between researchers and administrators

The system primarily benefits MSIRC administrators, staff, faculty members, and researchers. Administrators will have access to dashboards that allow them to monitor submissions, manage grant requests, track research progress, and generate reports efficiently. Researchers, on the other hand, will be able to submit proposals, upload documents, and check application statuses online, eliminating the need for repeated in-person transactions.

By addressing the challenges of manual documentation, fragmented records, and limited visibility of research activities, AYON provides a structured and functional solution that enhances efficiency, accountability, and transparency in research management within MSIRC-MSU

1.3 Objectives

This project entitled “AYON: Web- Based Integration Services Research Management System for Mamitua Saber Institute of Research And Creation (MSRIC) - MSU MAIN” aims to develop and implement a web-based system for managing research that will help MSRIC- MSU, its administrators, faculty members, and students to more effectively organize, manage, and access research-related information.

Specifically, this capstone project aims to

1. To gather and analyze the current processes of MSIRC-MSU in order to determine system requirements and specifications based on the role and activities of the end-user.
2. To develop the web-based management system based on the established requirements and specifications. aligned with the needs and functions identified from MSIRC-MSU administrators, staff, faculty, and researchers.
3. To design effective database structure for managing, organizing, and safeguarding research data, proposals, and completed research outputs.
4. To test and analyze to ensure that the developed system satisfies user requirements and adheres to the required specific criteria and procedures.

1.4 Scope and Limitations

This capstone project focuses on automating the core research services and internal processes of MSIRC in the MSU Main Campus. It covers processes involving the submission and monitoring of research proposals, research grant and incentive applications, research project review and approval, document validation and status tracking, digital storage, as well as a generating report to assist administrators in managing records better.

The project covers only the internal business of MSIRC, such as processing research proposals, Research Spotlight submissions, Travel subsidy Oral presentation requests, Internally funded research, Oral presentations, and Publication incentives. It specifically excludes externally funded research, external extension systems, and the Creative Division, which will be handled by other related researchers. Still, the system will implement a common database to ensure all data from MSIRC and its extensions are linked and consistent. The project also excludes financial management tasks, such as budgeting or accounting for research funds. User access within the system will be role-based, ensuring that only authorized personnel can perform specific tasks. Overall, this project focuses on improving documentation, monitoring, and management of research activities while excluding functions beyond these defined areas.

1.5 Significance of the Project

The proposed system AYON: Web- Based Integration Services Research Management System for Mamitua Saber Institute of Research And Creation (MSRIC) - MSU MAIN, is significant because it introduces innovation to the MSIRC's traditional workflow. It provides reliable and efficient solutions to existing administrative and documentation challenges. By reducing reliance on the paper-based processes, the system promotes environmental sustainability and aligns with the university's digitalization goals.

Faculty and Researchers. They can submit proposals and monitor the status of their applications online. This eliminates having to go to the office in person or sending printed forms, saving effort and time. With this ease, faculty and researchers can concentrate more on the quality and content of their studies.

College Dean, Research Coordinator, and Chairperson. These roles can monitor research progress, review outputs, and approve endorsements efficiently.

MSIRC Staff. Staff members can view submitted proposals, track their statuses, and assist with documentation or coordination.

MSIRC Special Assistant. The Special Assistant will have a faster and more organized way to review the completeness of submitted proposals and documents. The system allows them to validate files digitally, forward submissions efficiently, and reduce errors from manual checking.

MSIRC Director. The Director will have quicker access to pending proposals and can approve submissions online. The system generates automatic endorsement reports, making the review and approval process more organized and timely.

OVCRED. The Office of the Vice Chancellor for Research, Extension, and Development will receive endorsed proposals digitally, allowing them to issue final approvals and endorsements quickly. This improves coordination and ensures accurate tracking of approved research.

The Institution. strengthens the capacity to manage research outputs and track academic productivity. For administrators, it ensures smooth coordination and reduces manual workload. For faculty and researchers, it enhances accessibility to resources and improves the overall research experience.

CHAPTER 2

Review Of Related Literatures And Systems

The review is intended to provide useful background information for the researcher underpin the development of AYON: Web- Based Integration Services Research Management System for Mamitua Saber Institute of Research And Creation (MSRIC) - MSU MAIN. These concepts address the needs of Academic Institutions in Managing Research Activities. these

2.1 Review of Related Concepts

2.1.1 Document Management System (DMS)

Document Management Systems (DMS) are digital platforms used to store, organize, track, and manage documents electronically in a centralized environment. These systems support efficient document handling through features such as version control, access management, and workflow automation. In academic institutions, the adoption of DMS improves efficiency, accuracy, and accountability in managing administrative and academic records (Pagayonan, 2022).

Benavides et al. (2020) emphasized that digital transformation in higher education depends on the use of integrated systems that streamline documentation processes and reduce reliance on manual record-keeping. Similarly, studies on web-based document management systems for academic accreditation reported faster document retrieval, improved accuracy, and enhanced efficiency in handling institutional records (IJRES, 2021).

For the Mamitua Saber Institute of Research and Creation (MSIRC), implementing a document management system enables administrators, faculty members, and researchers to securely submit, access, and manage research documents electronically, reducing paperwork and improving overall workflow efficiency across the institute.

2.1.2 Web-Based Systems in Academic Institutions

Web-based systems are software applications accessed through web browsers, allowing real-time collaboration, remote accessibility, and integration with institutional databases. These systems simplify communication and information sharing among users regardless of location. In the education sector, adopting web-based platforms improves efficiency and accessibility in research and academic management (De Leon, 2020; Hernandez, 2021).

Benavides et al. (2020) emphasized that digital transformation in higher education relies on the integration of such platforms to enhance institutional operations and user experience. According to

Regla and Ballera (2023), implementing advanced web-based monitoring systems improves research productivity by enabling efficient tracking of faculty projects, even remotely.

For Mamtua Saber Institute of Research and Creation (MSIRC), a web-based design enables faculty, students, and administrators to manage research activities efficiently from any device connected to the Internet.

2.1.3 Database and Information Management

Databases form the backbone of any research or repository management system. They provide structured, secure, and retrievable storage of institutional data such as proposals, reports, and user information. Information management ensures data accuracy, reliability, and accessibility. The International Organization for Standardization (2020) outlines the principles of data security and integrity through the ISO/IEC 27001 standard, emphasizing confidentiality, availability, and protection of information assets.

For the MSIRC system, this ensures sensitive research data are well-protected and accessible only to authorized users through role-based controls. Information security is a vital component of all research management systems. Ratna et al. (2025) demonstrated how Role-Based Access Control (RBAC) can be implemented to manage user permissions efficiently and protect sensitive academic information from unauthorized access.

User authentication systems ensure that only authorized individuals can edit, approve, or view research materials. Mesa (2017) and Hernandez (2021) both recognized that password protection, secure logins, and user-level access design enhance confidentiality within academic systems. Advanced database management techniques, as described by Regla and Ballera (2023), further improve data tracking, analytics, and system responsiveness in higher education institutions.

2.1.4 Digital Repository and Archiving Systems

Digital repositories are online systems designed to store, organize, and preserve academic and research outputs such as theses, journal articles, and datasets. These systems play an essential role in ensuring long-term access and sustainability of scholarly materials. David and Alayon (2016) found that digital repositories in Philippine universities enhance research visibility and provide centralized access to institutional outputs. Integration of repository and archiving functionalities into an RMS ensures that completed research is not only stored securely but also remains discoverable for future studies.

2.2. Review of Related Systems

2.2.1 MSU-IIT Institutional Repository

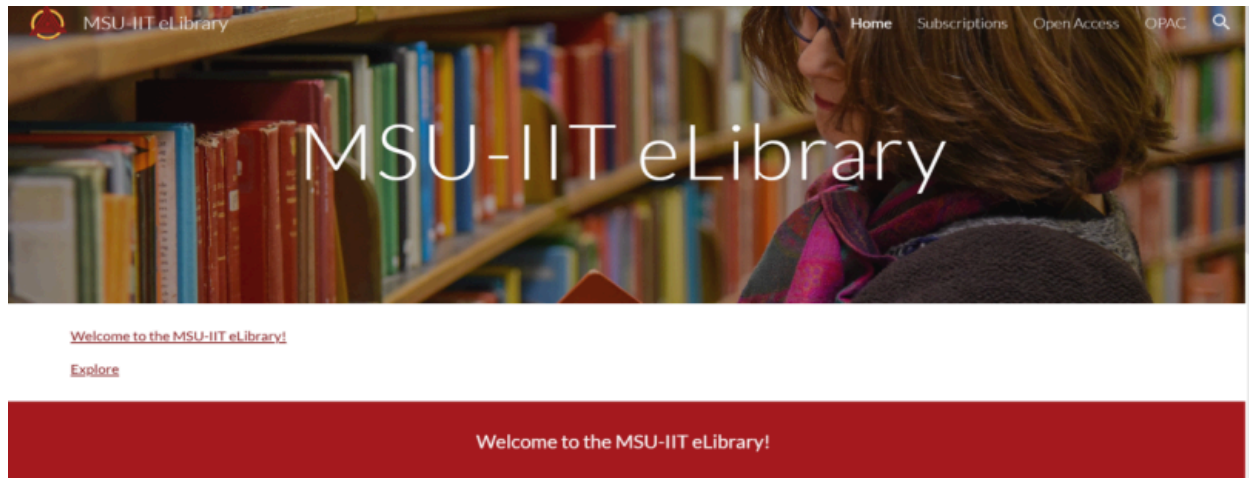


Figure 1: MSU-IIT Institutional Repository

The Mindanao State University–Iligan Institute of Technology (MSU-IIT) Institutional Repository serves as a digital archive designed to collect, preserve, and provide access to the scholarly works of the university community. It houses academic outputs such as undergraduate and graduate theses, dissertations, research publications, and other institutional documents. The system is managed by the MSU-IIT Library and aims to increase the visibility and accessibility of research materials produced by students and faculty members. It provides an online platform where users can browse and search for materials by author, title, department, or subject area. Additionally, the MSU-IIT eLibrary integrates digital resources and supports open access to a variety of research and academic materials.

The repository is significant because it helps maintain the intellectual legacy of the institution and supports the academic community in conducting research and referencing local studies. It follows metadata and cataloguing standards that allow easy retrieval of information. Despite its efficiency in managing institutional research outputs, some limitations are noted. The repository's coverage is mainly limited to the MSU-IIT campus, and there is no unified integration with other MSU campuses. Furthermore, features such as analytics dashboards, researcher profiles, and advanced metadata management are not yet fully implemented. These limitations highlight the need for a more comprehensive and system-wide research repository, which the proposed AYON: Web- Based Integration Services Research Management System for Mamitua Saber Institute of Research And Creation (MSRIC) - MSU MAIN. Intends to address.

2.2.2 TUKLAS: University of the Philippines Discovery Portal

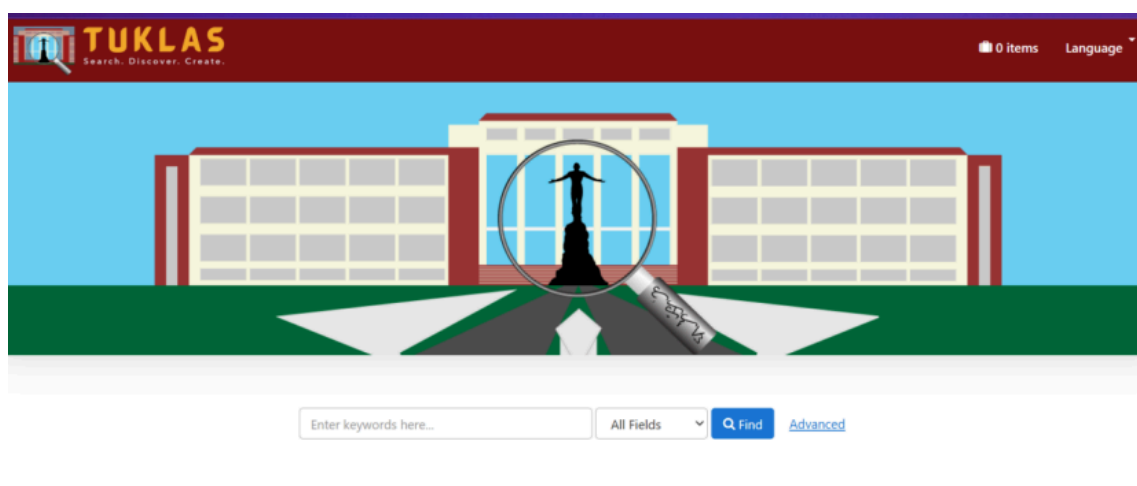


Figure 2: TUKLAS: University of the Philippines Discovery Portal

TUKLAS, also known as the University of the Philippines Discovery Portal, is a unified search platform that integrates the collections of all UP libraries into one system. It was developed to provide users with seamless access to both print and electronic academic resources across multiple campuses. TUKLAS allows users to search for books, journals, theses, dissertations, and other scholarly materials using a single interface. The system is built on open-source software and is customized to suit the needs of the UP academic community. It also includes features such as advanced search filters, citation generation tools, and access links to digital and physical collections. Through these features, TUKLAS simplifies the process of locating academic materials and promotes efficient research practices.

The significance of TUKLAS lies in its ability to centralize information from various UP campuses, thereby improving accessibility and collaboration among researchers and students. It represents a successful example of an integrated academic discovery system that unites different institutional repositories under one interface. However, TUKLAS also has its limitations. While it provides bibliographic information and access to metadata, it does not always grant full-text access to all materials. In addition, its focus is primarily on resource discovery and library integration rather than on managing research outputs or tracking research performance. These limitations demonstrate that while TUKLAS is a strong model for centralized access, AYON: Web- Based Integration Services Research Management System for Mamtua Saber Institute of Research And Creation (MSRIC) - MSU MAIN. Can expand its scope by including research management and data analytics capabilities for the MSU system.

2.2.3 DLSU Animo Repository



Figure 3:DLSU Animo Repository

The Animo Repository is the institutional repository of De La Salle University (DLSU), designed to collect, preserve, and provide access to scholarly and creative works produced by students, faculty, and researchers. It houses theses, research papers, institutional publications, and other academic outputs, supporting digital scholarship across the university ([DLSU Animo Repository](#)).

The repository is significant because it provides a centralized platform for archiving research outputs, ensuring proper metadata management and easy retrieval. Users can browse materials by department, author, or type of work, and access is provided through open access for authorized users. However, the system primarily focuses on archiving rather than research management. Features such as proposal submission, project tracking, and cross-campus integration are limited or absent. AYON: Web- Based Integration Services Research Management System for Mamtua Saber Institute of Research And Creation (MSRIC) - MSU MAIN. Can use the Animo Repository as a model for its repository module while extending functionalities to include workflow management, administrative dashboards, and system-wide integration.

2.2.4 Ateneo Research Portal / Archīum Ateneo



Figure 4: Ateneo Research Portal / Archīum Ateneo

Archīum Ateneo, the institutional repository of Ateneo de Manila University, collects and disseminates faculty, staff, and student research outputs. The portal provides access to research centers, ongoing projects, and publications, and supports submission of scholarly works through an authorized interface ([Archīum Ateneo](#); [Submit Research](#)).

The system is significant because it combines research visibility with repository functions, allowing centralized access to publications and project information. Users can search, browse, and submit research outputs, which promotes transparency and accessibility. However, Archīum Ateneo lacks comprehensive workflow management and cross-campus integration. AYON: Web- Based Integration Services Research Management System for Mamtua Saber Institute of Research And Creation (MSRIC) - MSU MAIN. Can adopt its structure for organizing outputs while integrating administrative dashboards, analytics, and multi-campus coordination to enhance research management.

2.3 Summary of Related Systems

The table below summarizes the key functions of the related systems. MSU-IIT Institutional Repository, TUKLAS, DLSU Animo Repository, and Ateneo Research Portal provide examples of platforms that manage research outputs, facilitate access to academic materials, and improve research visibility. These systems demonstrate features such as centralized data access, search and filter functions, user upload capabilities, and promotion of institutional research. These systems help guide the design of AYON: Web- Based Integration Services Research Management System for Mamtua

Saber Institute of Research And Creation (MSRIC) - MSU MAIN. by demonstrating useful features for research management and visibility.

Table 2.3: Summary of Related System

Function of Related System	MSU-IIT Institutional Repository	TUKLAS (UP)	Animo DLSU	Ateneo Research Portal / Archium Ateneo	AYON: MSU MAIN MSRIC (Proposed)
Digital Repository and Archiving of Research Outputs	✓	✓	✓	✓	✓
Centralized Access to Data	✓	✓	✓	X	✓
Search and Filter Functions	✓	✓	✓	✓	✓
Role-Based Dashboard and Access Control	X	X	X	X	✓
Research Visibility Promotion	✓	✓	✓	✓	✓
User Upload Function	✓	✓	X	✓	✓
SMS & Email Notification	X	X	X	X	✓

Research Grant and Incentive Application	X	X	X	X	✓
Report Generation	X	X	X	X	✓
Research Proposal Submission, Review, and Monitoring	X	X	X	X	✓

2.4 Technical Background

The Mamitua Saber Institute of Research and Creation (MSIRC) is the official research arm of Mindanao State University–Main Campus, located in Marawi City, Lanao del Sur. The institute was rebranded under Board of Regents (BOR) Resolution No. 368, s. 2024, and is dedicated to supporting the research and creative undertakings of the university’s faculty, staff, and students across various academic disciplines. Its mission is anchored on advancing instruction, research, and extension while preserving the rich socio-cultural and creative heritage of Mindanao.

The MSRIC serves as the central body responsible for formulating research policies, setting priorities, and evaluating research outputs within the university. It ensures that all research activities adhere to institutional standards and ethical guidelines. The institute also provides research and travel grants, monitors funded projects, and manages official university journals such as the Mindanao Journal and the ASEAN Journal for Social Sciences and Humanities (AJOSSH). Moreover, it serves as the main repository of research papers, creative works, and cultural materials that contribute to the university’s academic and creative excellence.

Before the establishment of a centralized digital system, the management and storage of research papers were done manually. Faculty and students submitted printed copies of their research works for approval and record-keeping, which often led to difficulties in tracking, retrieval, and preservation. These manual processes also increased the risk of data loss, misplacement, and limited accessibility to research materials.

The proponents aim to address these issues by developing a Mindanao State University Research Information and Creation System (MSRICS), a digital repository and management platform

designed specifically for the Mamitua Saber Institute of Research and Creation. Through this system, research proposals, outputs, and creative works will be stored, organized, and accessed digitally. The integration of this repository will ensure that all research information under the institute is efficiently managed, securely preserved, and easily retrievable.

By implementing the proposed MSRICS, the institute will enhance its ability to manage and safeguard research and creative outputs. This digitized approach supports the university's commitment to research excellence, innovation, and cultural preservation while ensuring that scholarly works remain accessible for future generations of researchers and creators.

2.4.1 Organizational Chart

The figure below shows the organizational chart of the Mamitua Saber Institute of Research and Creation (MSIRC) Mindanao State University–Main Campus,

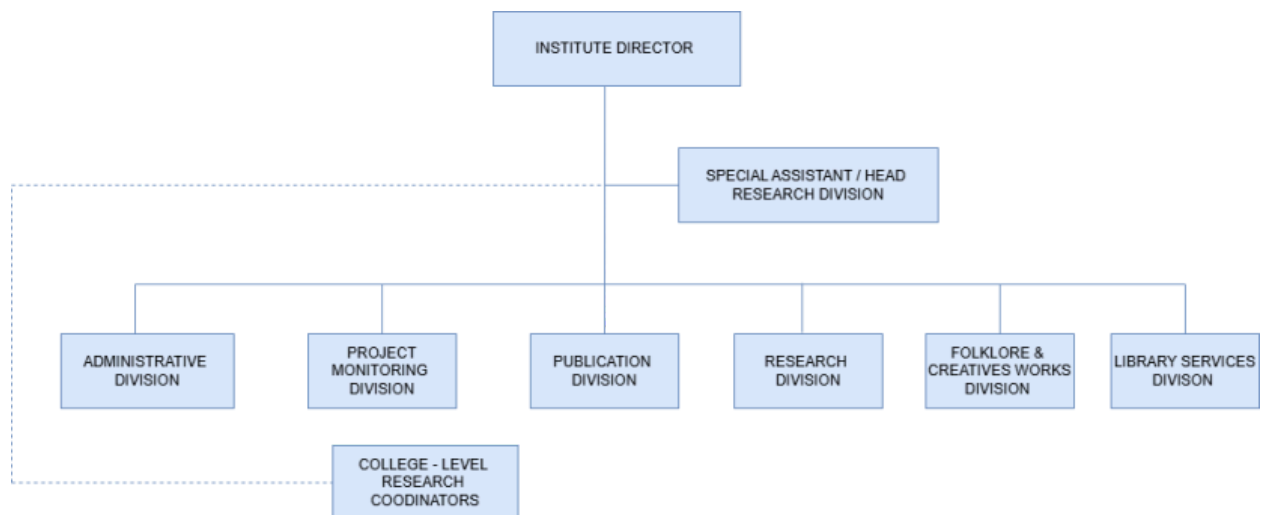


Figure 5: *Organizational Chart of Mamitua Saber Institute of Research and Creation (MSIRC) Mindanao State University–Main Campus,*

2.4.2 Workflow

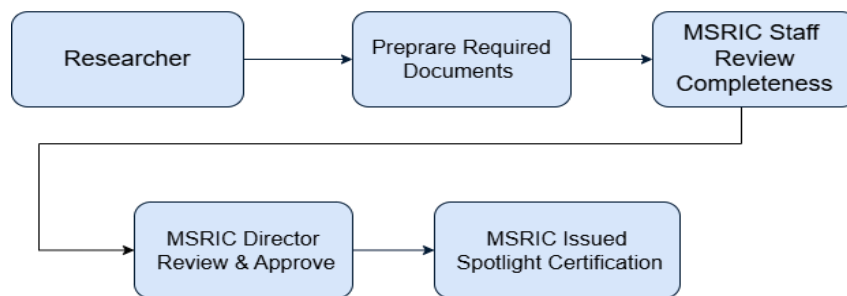


Figure 6: Existing Workflow Process University Research Spotlight

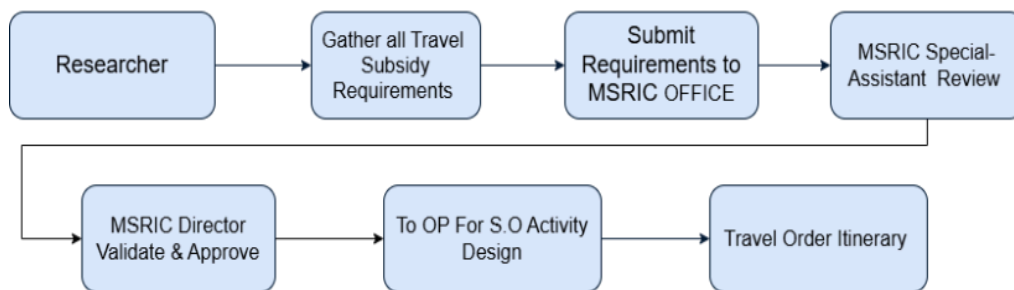


Figure 7: Existing Workflow Process for Travel Subsidy Oral Presentation

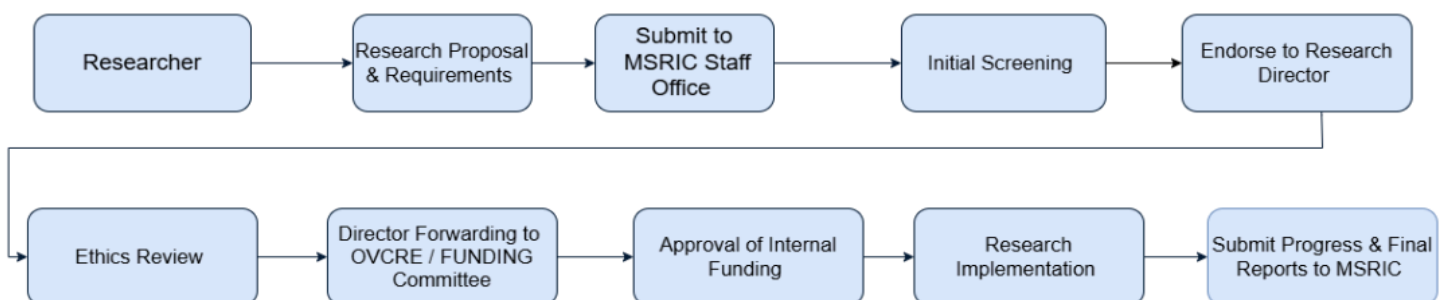


Figure 8: Existing Workflow Process Internally Funded Research

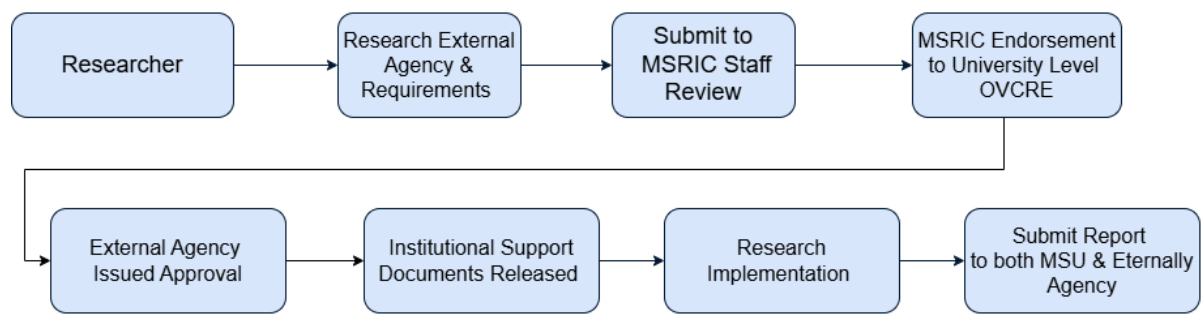


Figure 9: Existing Workflow Process of Externally Funded Research

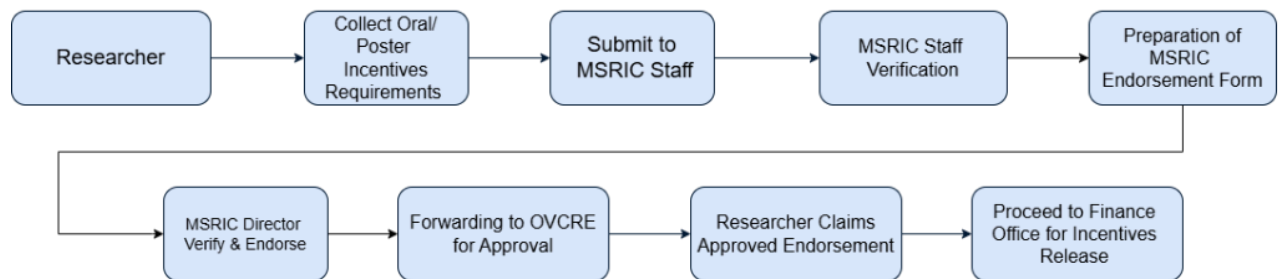


Figure 10: Existing Workflow Process of Oral Presentation Incentives

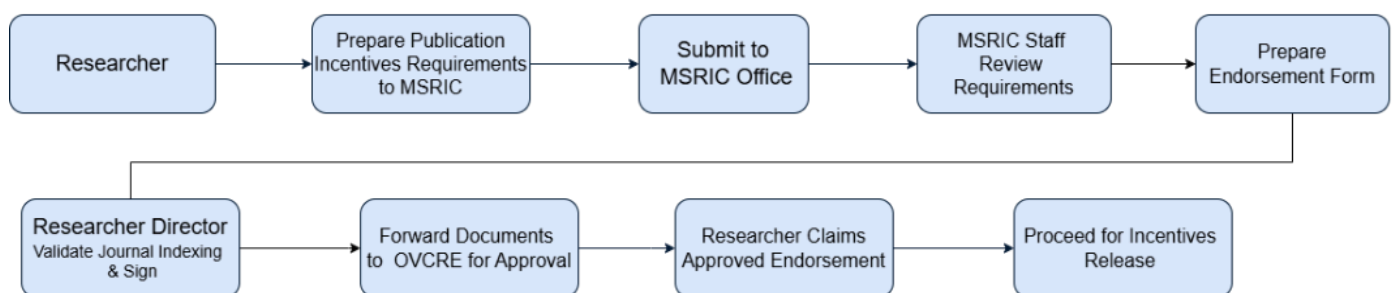


Figure 11: Existing Workflow Process of Publication Incentives

2.4.3 Proposed Workflow of the Proponent

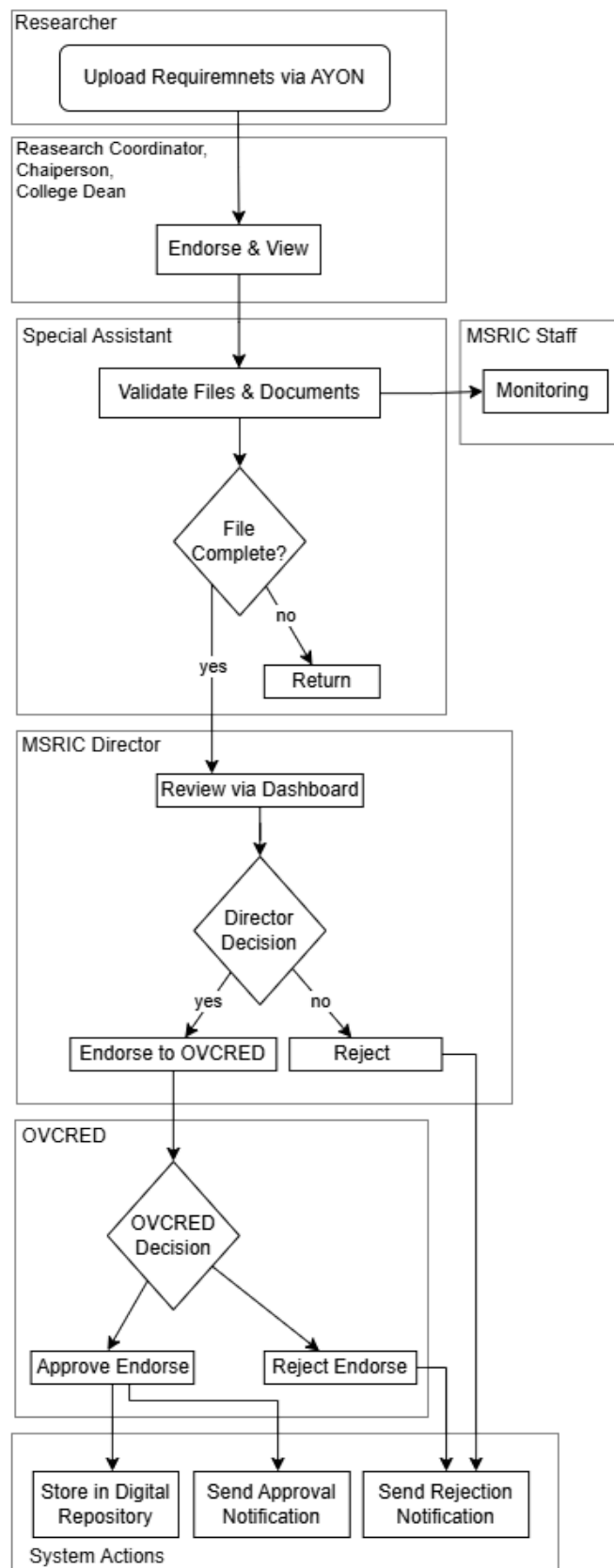


Figure 12: General Proposed Workflow Process

CHAPTER 3

Methodology

This chapter presents the methods and techniques that the researchers will use in conducting the study and in developing AYON: Web- Based Integration Services Research Management System for Mamitua Saber Institute of Research And Creation (MSRIC) - MSU MAIN. It explains the development methodology, data-gathering procedures, analytical tools, and the overall framework that will guide the creation of the system.

To support an organized, flexible, and user-centered development process, the researchers will adopt the Agile “Scrum” Methodology. This methodology will allow the team to work in iterative cycles and to make continuous improvements based on stakeholder feedback. Since MSRIC’s operations involve evolving workflows and multiple users, Agile Scrum will help ensure that the system develops gradually in manageable phases while meeting the actual needs of MSIRC staff, faculty, and researchers.

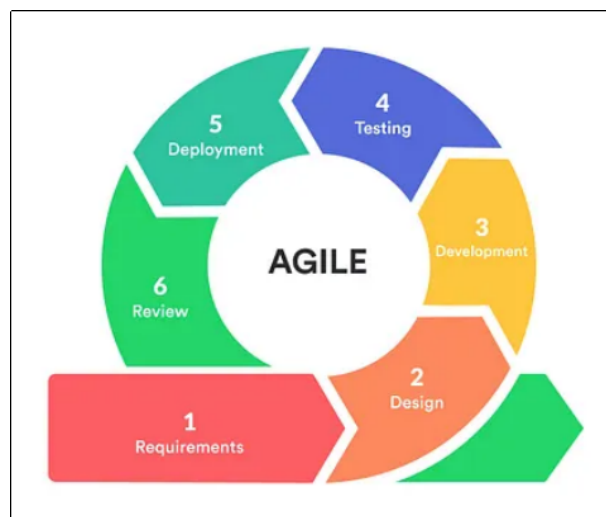


Figure 3.1 Agile “Scrum” Methodology

3.1 Requirements Gathering

Data gathering methods such as interviews, observations, and document reviews will be conducted to understand MSRIC’s existing research management procedures. These activities will help identify problems in the current manual workflow, including delays in proposal processing, tracking difficulties, and challenges in maintaining research records. The information collected during this stage will guide the system’s analysis, design, development, and evaluation.

3.1.1 Empathy Map

SAYS	FEELS
"I need to submit my proposal before the deadline." "Where can I check my application status?" "I want to apply for research incentives." "How do I access past research outputs?"	Frustrated with delays in approval process Uncertain about document completeness Hopeful for quick approval and incentives Relieved when receiving SMS notifications

Figure 3.1.1.1 Empathy Map of Researchers

Figure 3.1.1.1 presents the Empathy Map for Researchers, representing faculty members and students who submit research proposals and grant applications. The map captures what researchers Say, Feels, Think and Does in their current research workflow. The Empathy Map guided the design of user-friendly features such as real-time tracking, online document submission, and notification systems.

Figure 3.1.1.2 Empathy Map of Special Assistant

Figure 3.1.1.2 shows the Empathy Map for the Special Assistant, who plays a central role in validating submitted documents and forwarding proposals for review. This Empathy Map supports system features such as automated completeness checks, digital validation tools, and a streamlined dashboard for efficient file forwarding.

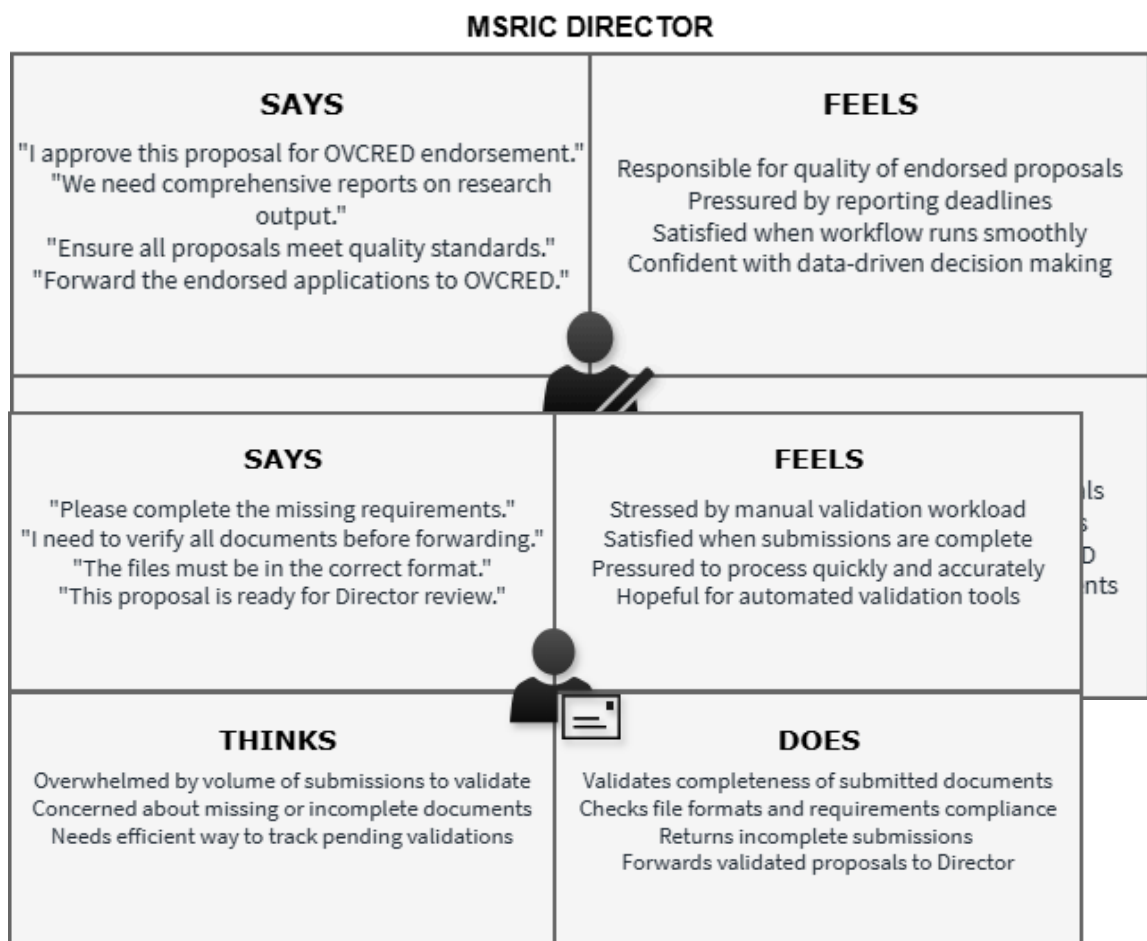


Figure 3.1.1.3 Empathy Map of Director

Figure 3.1.1.3 presents the Empathy Map of the MSRIC Director, showing the Director’s core responsibilities and experience in handling research proposals. This map highlights the Director’s need for a streamlined, well-organized workflow.

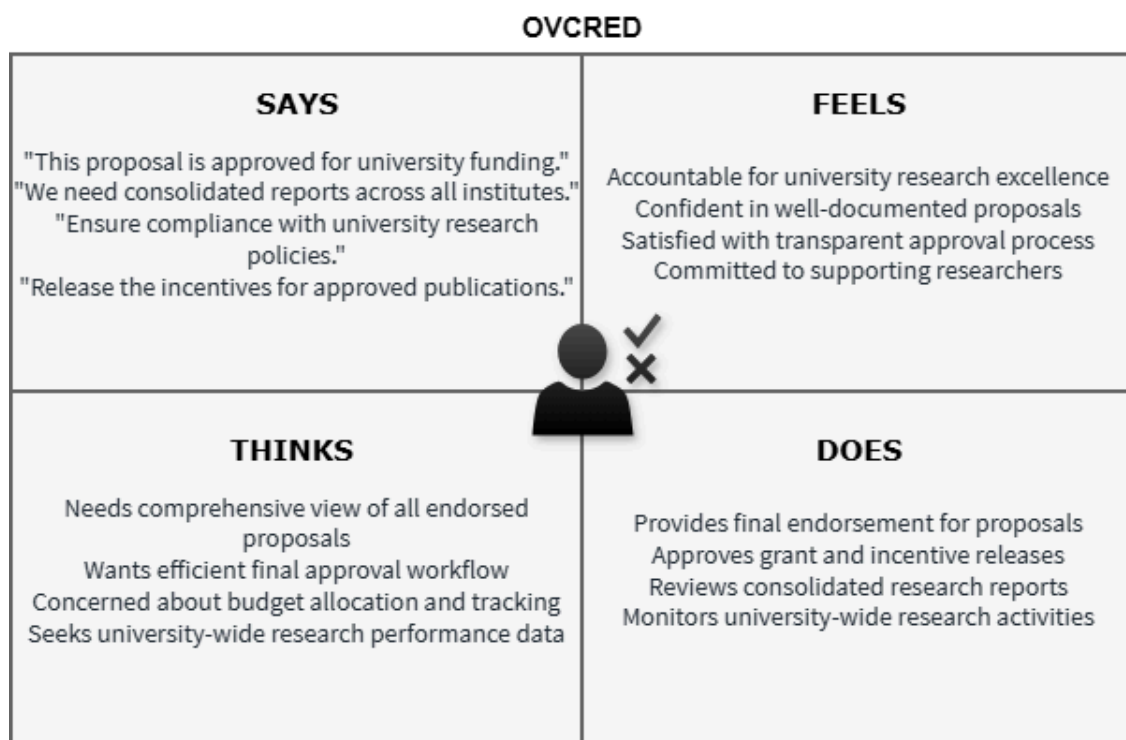


Figure 3.1.1.4 Empathy Map of OVCRED

Figure 3.1.1.4 illustrates the Empathy Map for the Office of the Vice Chancellor for Research, Extension, and Development (OVCRED). The Empathy Map highlights the need for a centralized approval system, automated routing, and digital endorsement features to speed up final approval stages.

3.1.2 Functional Requirements and Non-functional Requirement

Functional Requirements define what the system should do. The following table outlines the specific functionalities that the "AYON: Web- Based Integration Services Research Management System for Mamitua Saber Institute of Research And Creation (MSRIC) - MSU MAIN", Must

provide to meet the needs of MSRIC. While the Non-Functional Requirements define the utility attributes and constraints of the system. These requirements specify how the system should perform rather than what it should do.

Table 3.1.2 Functional and Non-functional Requirement

Requirements	Type	Priority
System shall allow users to register, login, and manage their accounts with role-based access	Functional	Mandatory
System shall allow researchers to submit and track research proposals online	Functional	Mandatory
System shall enable researchers to apply for grants and incentives and track application status	Functional	Mandatory
System shall allow administrators to review, approve or reject research proposal online	Functional	Mandatory
System shall provide a searchable online library where completed research works can be viewed.	Functional	Mandatory
System shall automatically generate reports about proposals, grants, and research activities.	Functional	Mandatory
System shall provide a role-based access control for users to maintain security and data privacy	Functional	Mandatory
System shall notify researcher via SMS when their request is approved or rejected	Functional	Mandatory
System shall load pages within 5 seconds under normal network conditions	Non-Functional	Optional
System shall be available 99.5% uptime during operational hours	Non-Functional	Mandatory
System shall have an intuitive interface requiring minimal training for users	Non-Functional	Optional
System shall be compatible with major browsers (Chrome, Firefox, Mozilla, Edge)	Non-Functional	Mandatory

3.1.3 Gantt Chart

Table 3.1.3 Gantt Chart

	2025								2026													
	Sept - Oct				Nov - Dec				Jan - Feb				Mar - Apr				May - Jun					
Phase/Task	w 1	w 2	w 3	w 4	w 1	w 2	w 3	w 4	w 1	w 2	w 3	w 4	w 1	w 2	w 3	w 4	w 1	w 2	w 3	w 4		
Chapter 1-2: Introduction & Literature Review																						
Introduction																						
Literature Review																						
Chapter 3: Methodology																						
Requirement Analysis																						
System Design																						
Development Phase																						
Database Setup																						
Authentication Module																						
Proposal Submission Module																						
Grant & Incentive Application																						
Review & Approval Workflow																						
Report Generation Module																						
Notification System																						
Testing Phase																						
Unit Testing																						
Integration Testing																						

3.2 Design

3.2.1 Hierarchical Input-Process-Output (HIPO)

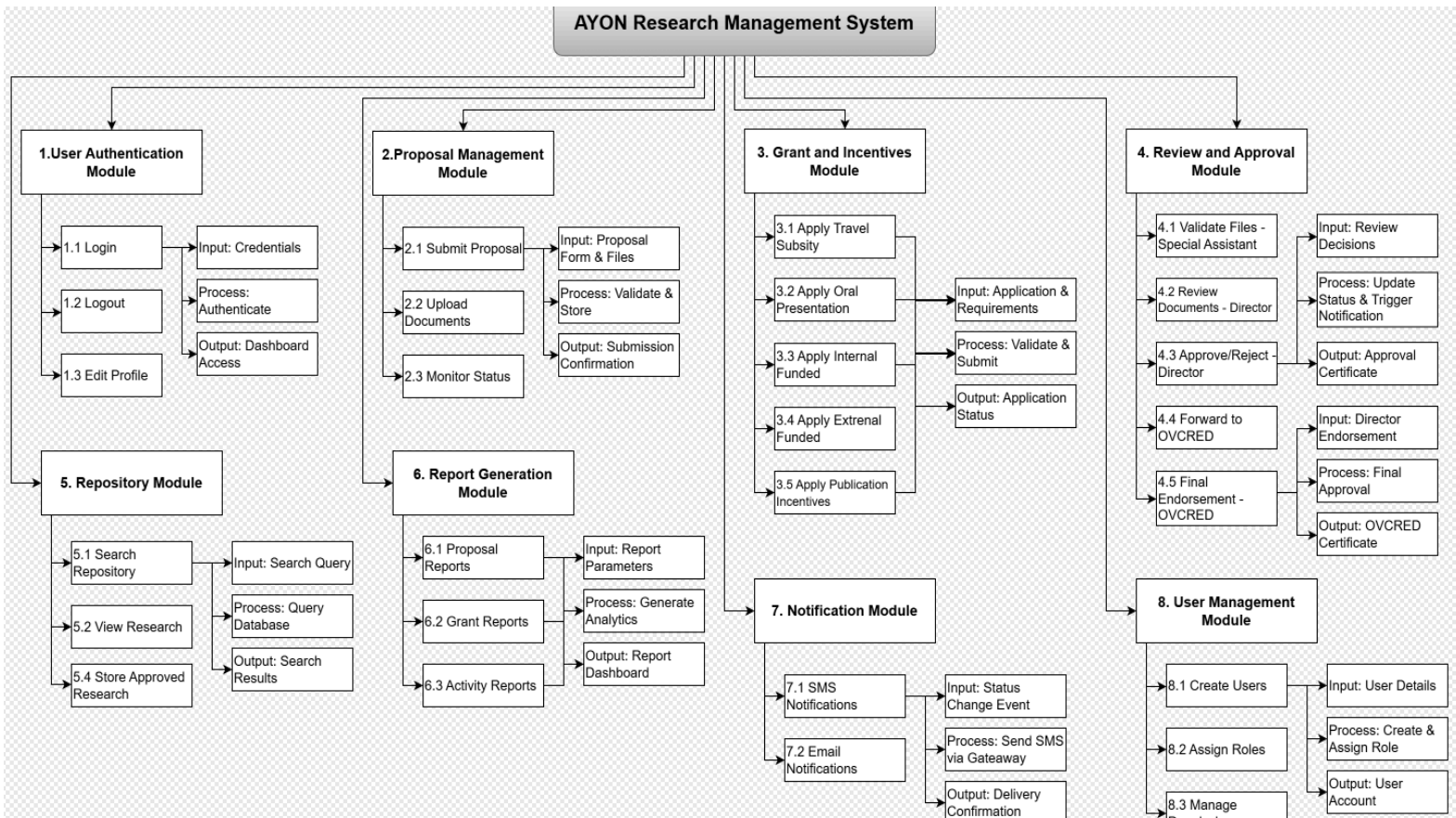


Figure 3.2.1 Hierarchical Input-Process-Output

Figure 3.3.1 shows The Hierarchical Input-Process-Output (HIPO) diagram provides a visual representation of the system's modular structure. It breaks down the AYON Web-Based Research Management System into eight main functional modules, each corresponding to the key features identified in the requirements analysis. The diagram shows how each module is further decomposed into sub-functions, with their respective inputs, processes, and outputs. This hierarchical approach supports the system's role-based architecture, where researchers interact primarily with proposal submission and grant application modules, the Special Assistant handles file validation and forwarding, the MSIRC Director manages review and approval processes, and OVCRED provides final endorsement for approved proposals. The modular design ensures maintainability, scalability, and clear separation of concerns.

3.2.2 Input-Process-Output (IPO)



Figure 3.2.2 Input-Process-Output

Figure 3.3.2 Shows The Input-Process-Output (IPO) diagram illustrates the data flow throughout the AYON system. It shows how various inputs from different actors are processed and transformed into meaningful outputs. Researchers provide credentials, proposal documents, and application forms as inputs. The Special Assistant validates these submissions and forwards complete files for review. The MSIRC Director provides review decisions and generates endorsement forms. OVCRED provides final approval decisions for proposals endorsed by the Director. The system processes these inputs through authentication, validation, storage, review workflows, and notification services, ultimately producing outputs such as user dashboards, submission confirmations, approval certificates, OVCRED endorsements, searchable research repositories, and automated SMS/email notifications.

3.2.3 Use Case Diagram

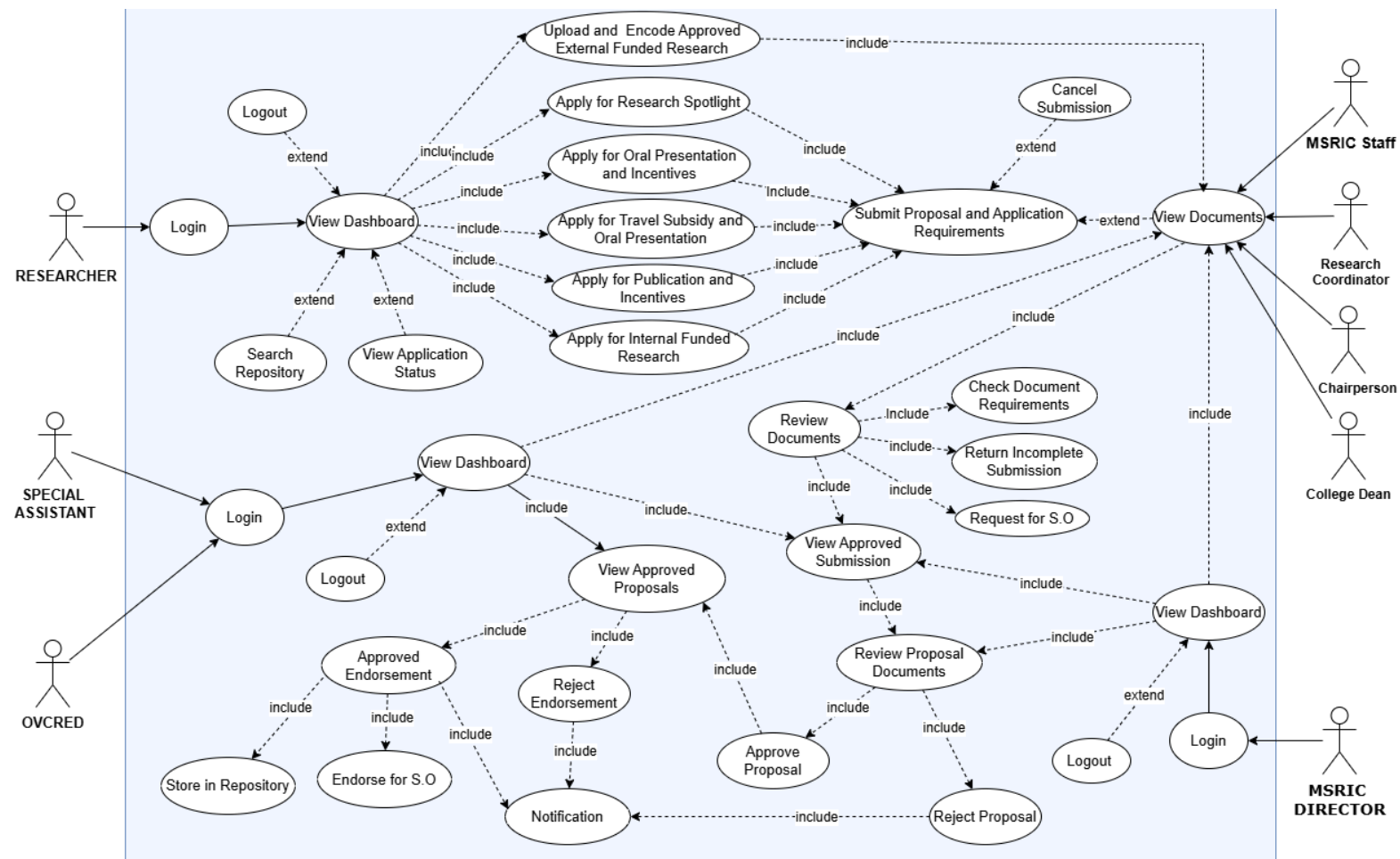


Figure 3.2.3.1 Use Case Diagram

Figure 3.2.3 The Use Case Diagram represents all the functional requirements of the AYON Web-Based Research Management System and illustrates how the four primary actors interact with the system. The diagram is organized into functional subsystems that correspond to the key features identified in the requirements analysis.

3.2.4 Entity-Relational Diagram

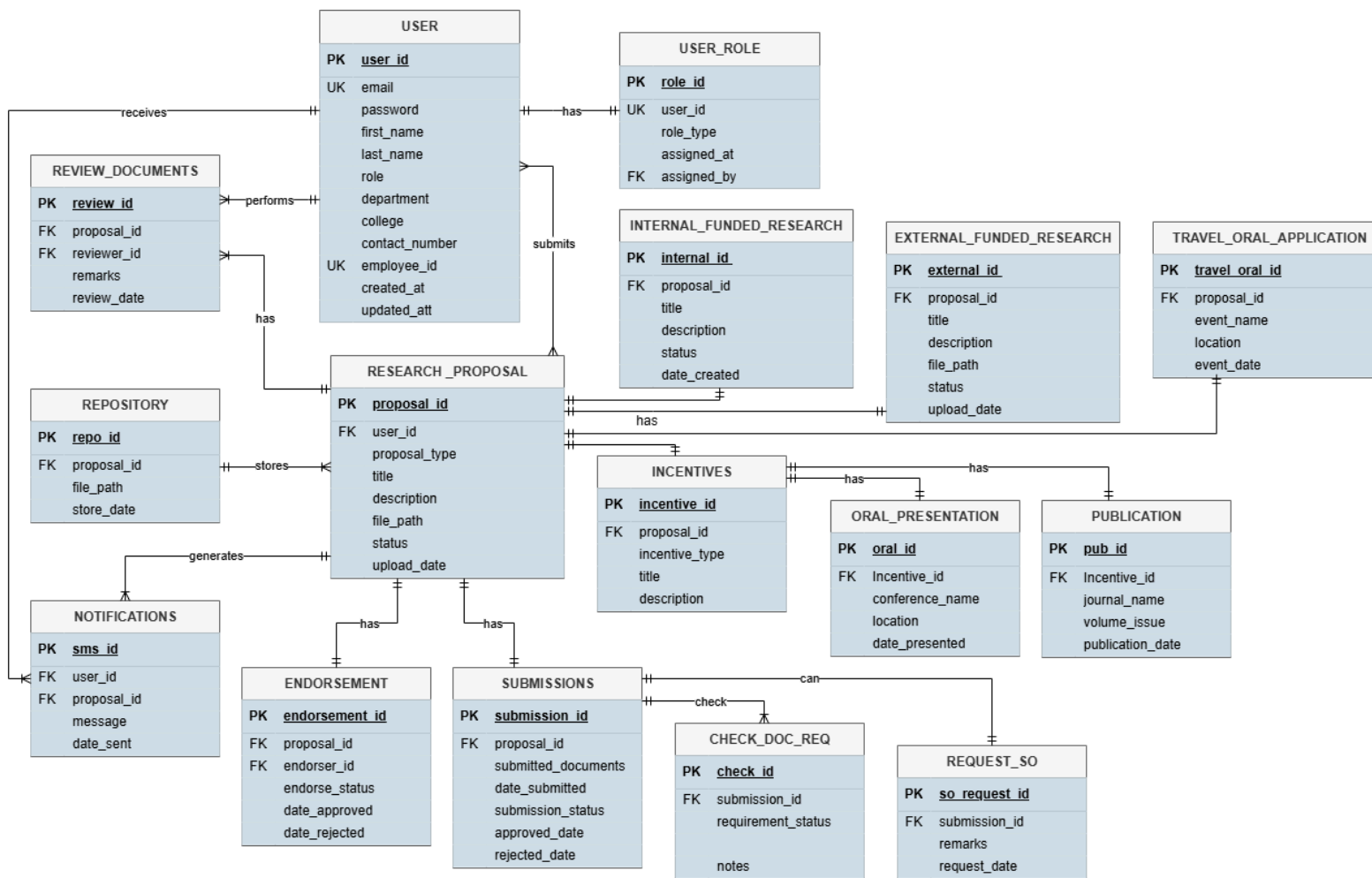


Figure 3.2.4 The Entity-Relationship Diagram (ERD)

Figure 3.2.4 The Entity-Relationship Diagram (ERD) represents the database structure of the AYON system, showing all entities, their attributes, and the relationships between them. The diagram is derived from the Use Case Diagram and reflects the data requirements for all four actors: Researchers, Special Assistant, MSIRC Director, and OVCRED.

3.2.5 Architectural Design

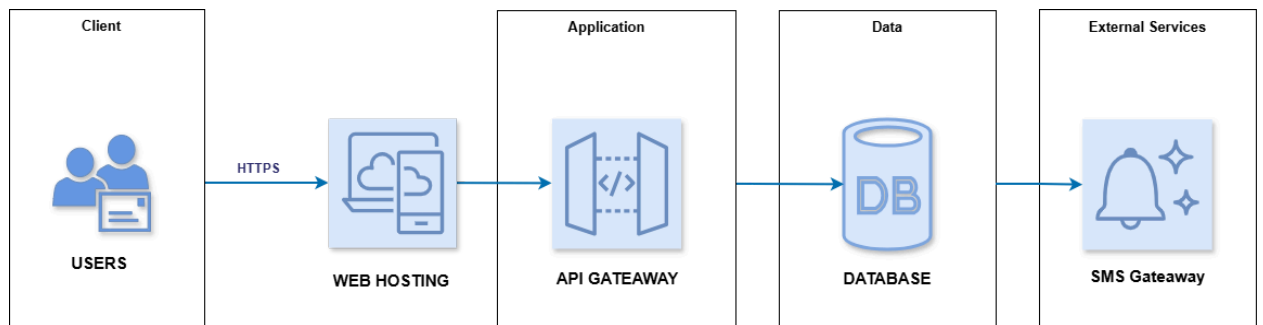


Figure 3.2.5 The Architectural Design

Figure 3.2.5 The Architectural Design follows a 3-tier architecture pattern that separates concerns across the client, application, and data layers. This design supports the multi-actor workflow involving Researchers, Special Assistant, MSIRC Director, and OVCREd, ensuring each actor has a dedicated dashboard interface while sharing common backend services.

3.3 Development

3.3.1. Software Specification

Table 3.3.1.1 Developer's Software Requirements

Components	Specifications
Operating System	Windows 10/11
DBMS	MySQL, Oracle
Web browser	Google Chrome, Mozilla FireFox, Microsoft Edge
Programming Language	HTML, CSS, JavaScript, React .js (Frontend) PHP/Node. Js (Backend)

Programming Environment	Visual Studio Code
Document Editor	draw.io

Table 3.3.1.2 User’s Software Requirements

Components	Specifications
Operating System	Windows 10 or 11
Web Browser	Google Chrome, Mozilla Firefox, Microsoft Edge
Internet Connection	Broadband connection (minimum 2 Mbps recommended)
PDF Reader	Adobe Acrobat Reader or Browser built-in PDF viewer

3.3.2. Hardware Specification

Table 3.3.2.1 Developer’s Hardware Requirements

Component	Specifications
Processor	Intel Core i5/i7
RAM	8 GB minimum (16GB recommended)
Storage	128 GB minimum (256GB recommended)
Monitor Resolution	1024 x 768 minimum (1920 x 1080 recommended)

Input Devices	Standard Keyboard and Mouse
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Table 3.3.2.2 User's Hardware Requirements

Component	Specifications
Processor	Intel Core i5 minimum (Intel Core i7 recommended)
RAM	4 GB minimum (8-16 GB recommended)
Storage	128 GB minimum (256GB recommended)
Display	1024 x 768 resolution
Network	3G/LTE or Wi-Fi

3.4 Testing

3.4.1 Respondents

The system will be evaluated by respondents representing the four primary user roles (actors) who will directly interact with the AYON system.

- Researcher - Faculty members or students who submit research proposals, apply for grants/incentives, upload research outputs, and track proposal status
- Special Assistant - MSIRC staff responsible for initial document validation, file completeness checking, and forwarding proposals to the Director.
- MSRIC Director - Reviews and evaluates proposals for content quality, alignment with institutional priorities, and provides recommendations to OVCRED
- OVCRED - Office of Vice-Chancellor for Research and Development provides final approval/rejection.

3.4.2 Procedure

The testing procedure will follow a systematic approach to ensure thorough evaluation of the system:

- **System Demonstration:** A comprehensive walkthrough of all system features will be presented to the respondents.
- **Hands-on Testing:** Respondents will be given access to test accounts with their respective roles (Researcher, Special Assistant, Director, OVCRED).
- **Task Completion:** Users will perform specific tasks such as submitting proposals, validating documents, and approving submissions.
- **Questionnaire Distribution:** After testing, respondents will be given a System Usability Scale (SUS) questionnaire to answer.
- **Data Collection:** All responses will be collected and tabulated for statistical analysis.
- **Feedback Session:** An open forum will be conducted to gather qualitative feedback and suggestions for improvement.

3.4.3 Tools (Questionnaire)

The database application system will be tested and evaluated using SUS (System Usability Scale); it is a reliable tool for measuring usability. It consists of 10 questionnaires rated on a five-point Likert scale; from Strongly Agree to Strongly Disagree. Originally created by John Brooke in 1989, it will evaluate this capstone project.

		Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
		1	2	3	4	5
1	I think that I would like to use this system frequently.	☐	☐	☐	☐	☐
2	I found the system unnecessarily complex.	☐	☐	☐	☐	☐
3	I thought the system was easy to use.	☐	☐	☐	☐	☐

4	I think that I would need the support of a technical person to be able to use this system.	☐	☐	☐	☐	☐
5	I found the various functions in this system were well integrated.	☐	☐	☐	☐	☐
6	I thought there was too much inconsistency in this system.	☐	☐	☐	☐	☐
7	I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☐
8	I found the system very cumbersome to use.	☐	☐	☐	☐	☐
9	I felt very confident using the system.	☐	☐	☐	☐	☐
10	I needed to learn a lot of things before I could get going with this system.	☐	☐	☐	☐	☐

Table 3.4.3: SUS Item Questionnaire

The calculation process of the SUS score starts by summing the score contribution from each item. Each item's score contribution ranges from 0 to 4. For items 1, 3, 7, and 9 (the positive worded items), the score contribution is the scale position minus 1. For items 2, 4, 6, 8 and 10 (the negatively worded items), the contribution score is 5 minus the scale position. To obtain the overall value of SUS, the sum of the score of all items is taken and then multiplied to 2.5 the SUS scores range between 0 to 100 (Brooke, 1986).

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